ELIM AVPRO + Wing Build: Pack for Matt

Generated from the build set on 2026-09-28 (BOM v7). Don't edit this file; it's rebuilt from `build/`.

1. Scope

ELIM Church runs a portable production rig that deploys once or twice a week at different venues, set up and struck by volunteers under time pressure. This build consolidates the video, network and control gear into a ProX AVPRO rolling workstation (two 12U bays, two 4U tops, a work surface and two monitor lids). Bay A is the active-system side of the rack; Bay B is the storage and camera-charging side. The Behringer Wing rides in a ProX BWING flip case beside it. Every field cable (cameras, LED wall, confidence monitor, internet) and the five-cable Wing umbilical lands on one labeled, color-coded Neutrik D-series panel on the back of the AVPRO. Almost all the gear inside is already owned. The new spend is the cases, power, cooling, mounts, connectors, cabling and two lid monitors.

The design is locked. We'd like your quote on the cases and freight, your help pinning a few connector parts, and a sanity check from an outside AV pro.

2. Questions for you

1. **Case and freight quote:** AVPRO price, BWING price, consolidated freight for both (closed AVPRO is 51.5 × 48.2 × 31.6in). BOM placeholders: $500 / $900 / $250
   When: 2026-10-01. Gates the order? Yes: the order gate
2. **One matched powerCON TRUE1 family:** AVPRO chassis outlet, both umbilical cable ends, and the Wing chassis inlet (only if the Wing panel happens). **Plus the etherCON pair:** NE8FDP-B panel jacks with NE8MX-1 carriers (standard etherCON; the NE8MC-1 Sam named is discontinued and the NE8MX-1 is its successor), checked against the Neutrik compatibility table before the BOM calls them firm
   When: 2026-10-01. Gates the order? Yes, for the connector lines
3. **Rack rail thread:** tapped 10-32, or square-hole for M6 cage nuts? Assumed 10-32. Do you know? If not, we check the rails at delivery
   When: on case delivery (assumed by 2026-11-15). Gates the order? Screw line only. If you don't know, we order the screws after delivery
4. **Laptop power strip:** a plain surge strip for the desk laptops, fed from the utility side (Bay B), never behind the UPS. BOM carries $30
   When: 2026-10-01. Gates the order? That line only
5. **BWING inspection: can it take a panel-mount D-series plate?** Yes → buy the Wing block. No → umbilical direct to the Wing's jacks + a TRUE1-to-IEC jumper
   When: on case delivery (assumed by 2026-11-15). Gates the order? Wing block only
6. **Side doors:** do the AVPRO side doors open to usable storage, and what are the inside dimensions? The design doesn't use them
   When: on case delivery (assumed by 2026-11-15). Gates the order? No
7. **The "regret ask":** long-lead items, freight consolidation, anything hard to add later
   When: 2026-10-01 (with the quote). Gates the order? No

3. Panel maps

AVPRO rear I/O panel

Three Neutrik D-series 12-hole plates (3U) on the Bay A rear rails, behind the upper gear block. This is the one face volunteers touch. Field cables land here; short jumpers run to the gear. Nothing plugs into a device directly. *Decided.*

```
PANEL 1  [CAM1][CAM2*][CAM3][CAM4][CAM5][CAM6][CAM7][CAM8*][CONF OUT][ -- ][ -- ][ -- ]
PANEL 2  [DANTE][LED1][LED2][INTERNET IN][PROD][TALKBACK][IEM][SPARE][ -- ][ -- ][ -- ][POWER > WING]
PANEL 3  [ spare / expansion row, 12 holes ]
         * shared jack (see the signal map)   -- = blanking plate
```

Jack	Connector	Direction / gender	Inside, goes to	Color
CAM 1, 3–7	BNC 75Ω feedthrough	in	SDI jumper → ATEM IN n	blue
CAM 2, CAM 8	BNC 75Ω feedthrough	in, shared	jumper parked at ATEM IN 2 / IN 8	blue
CONF OUT	HDMI feedthrough	out	Anker hub HDMI	white
DANTE TRUNK	etherCON	both	satellite switch	green
LED 1, LED 2	etherCON	out	LED transmitter ports 1, 2	purple
INTERNET IN	etherCON	in	AX21 WAN	red
PROD TRUNK	etherCON	both	AX21 LAN 3	yellow
TALKBACK	XLR **female**	in (a mic plugs in)	trunk to the Wing	orange
IEM	XLR **male**	out (feeds a pack)	from the Wing	orange
SPARE	etherCON	both	AX21 LAN 4 (pre-wired)	yellow
POWER > WING	powerCON TRUE1 **outlet**	out	fed from the CPS1215RM (UPS side)	black

- Order and labels follow layout v15 (POWER > WING last; TRUNK names). *Decided.*
- The POWER jack is an **outlet**: power leaves the AVPRO here. Shore power enters through the UPS line cord, not this jack.
- 18 empty holes get blanking plates. Panel 3 is the growth row (camera returns, more trunks).

Wing panel (conditional)

Planned, pending physical inspection of the BWING. Nobody has confirmed the BWING can take a panel. Sam inspects on case delivery. The parts are a separate BOM block ($100.00) and are not in the first order. *Conditional.*

```
WING PANEL  [CONTROL][DANTE TRUNK][AES50][TALKBACK][IEM][POWER IN][ -- ][ -- ]
```

Jack	Connector	Gender / direction	To
CONTROL	etherCON	both	Wing control port → PROD trunk → AX21
DANTE TRUNK	etherCON	both	Wing case Dante switch uplink
AES50	etherCON	both	stage box (stage side, not on the umbilical)

Jack	Connector	Gender / direction	To
TALKBACK	XLR **male**	out to the Wing In 8	mirrors the AVPRO's female
IEM	XLR **female**	in from the Wing Out 8	mirrors the AVPRO's male
POWER IN	powerCON TRUE1 **inlet**	in	pigtail to the Wing's IEC inlet

- **Inside the case:** the Wing case Dante switch (new 8-port, TL-SG108 or equivalent) and Tyler's Mac. Tyler's Mac has two links to the Wing: USB-C for recording and a Cat patch to the Dante switch.
- Genders mirror the AVPRO panel: talkback female ↔ male, IEM male ↔ female, POWER outlet ↔ inlet.
- **Fallback if the BWING can't take a panel:** the umbilical runs direct to the Wing's own jacks. etherCON ends plug into ordinary RJ45 jacks (Wing control, the Dante switch). The XLRs plug into Wing In 8 / Out 8. Power: the umbilical's female TRUE1 end feeds the Wing through a short TRUE1-to-IEC C13 jumper (BOM fallback line, qty 0 until needed).

4. Port and signal map

Signal map

Video

From	Cable	To	Color
Wired Pocket 4K (converter)	SDI field, 125ft	Panel CAM 1 → jumper → ATEM IN 1	blue
Hollyland RX (4U top)	SDI, short	ATEM IN 2 direct	blue
DJI (converter)	SDI field, 125ft	Panel CAM 3 → jumper → ATEM IN 3	blue
Rented cameras	SDI field	Panel CAM 4–7 → ATEM IN 4–7	blue
ProPresenter Mac → UltraStudio Monitor (desk)	USB-C, then SDI	ATEM IN 8 direct	blue
ATEM AUX	SDI	HyperDeck HD Mini (records the clean feed)	blue
ATEM OUT 8	SDI	Resi encoder	blue
ATEM multiview	HDMI	LEFT lid monitor	white
ProPresenter Mac, native HDMI	HDMI	LED wall transmitter (Bay A U9)	white
LED transmitter, 2 × ethernet	Cat, short	Panel LED 1, LED 2 → LED wall	purple
ProPresenter Mac → Anker hub	HDMI	Panel CONF OUT → confidence monitor (stage)	white
RIGHT lid monitor	—	fed in Phase 3 (Mac Minis, dock or splitter)	—

Shared jacks (CAM 2, CAM 8). The Hollyland RX and the UltraStudio plug straight into ATEM IN 2 and IN 8. The CAM 2 and CAM 8 panel jumpers are run to the desk and parked beside those inputs. To use either jack for a field camera, swap the jumper in at the ATEM. Label both jacks "shared, see ATEM IN 2 / IN 8". *Decided.*

Audio and Dante

From	Cable	To	Color
Stage box (Midas DL251)	AES50	Wing (via the Wing panel AES50 jack if the panel happens)	orange boot

From	Cable	To	Color
Wing Dante card	Cat	Wing case Dante switch (new 8-port, TL-SG108 or equivalent)	green
Wing case Dante switch	DANTE trunk	AVPRO panel DANTE TRUNK → Bay A Dante satellite (TL-SG1005P)	green
Satellite switch	Cat, PoE	AVIO (desk). The AVIO is PoE-powered	green
AVIO	XLR pigtails	ATEM audio Ch 1–2	orange
Bianca's Mac, USB-C ethernet adapter	Cat	satellite switch (Dante)	green
Stream-mix laptop (Omar/Mitch, in the booth)	Cat	Wing case Dante switch	green
Tyler's ProTools Mac (in the Wing case)	Cat	Wing case Dante switch. It's a Dante endpoint as well as the USB-C recorder	green
Bianca's talkback mic	XLR	AVPRO panel TALKBACK → trunk → Wing XLR In 8	orange
Wing XLR Out 8	XLR trunk	AVPRO panel IEM → Bianca's IEM pack	orange
Wing	USB-C	Tyler's Mac (in the Wing case) for recording	—

- Wing In 8 / Out 8 come from the current rig's port map. **Tyler confirms both.** Until then they're "to be confirmed by Tyler".

Production network and internet

From	Cable	To	Color
Venue drop or Starlink	Cat field	Panel INTERNET IN → AX21 WAN	red
AX21 LAN 1	Cat	Production switch (TL-SG105)	yellow
AX21 LAN 2	Cat	Resi	yellow
AX21 LAN 3	PROD trunk	Panel PROD TRUNK → Wing CONTROL	yellow
AX21 LAN 4	Cat	Panel SPARE (etherCON, pre-wired)	yellow
Production switch	Cat	ATEM, HD Mini (auto-trigger), switcher laptop, Bianca's Anker hub	yellow

- **Port count:** that fills every wired production port (AX21 4 of 4, switch 5 of 5). A future device (Stream Deck host, Mac Mini) needs a bigger switch (a bigger production switch is on the roadmap).

5. Power

Power

The rig runs on whatever a venue gives. It draws well under one 15A circuit, so one works. **Two are preferred:** the UPS on one, the charging side on the other. Never tie circuits together. *Decided.*

Venue power drops

Power reaches the rig area on **two 125ft 12/3 SOOW drops** from wall outlets. *Decided.*

Drop	Wall end	Rig end	Feeds
Drop 1	5-15P plug	one 5-15R connector body	the AVPRO UPS line cord (and the Wing, through the AVPRO)
Drop 2	5-15P plug	one 5-15R connector body	the AV charging station: the Bay B CPPDU511 cord direct (Sam's leaning), or a quad box only if the station is a separate table (tentative)

- **Both drops are built the same:** 125ft 12/3 SOOW, a Hubbell HBL5266C plug, a Hubbell HBL5269C connector body. Either one can stand in for the other. The Hubbell part numbers are assumed, not a ruling; an equal 15A industrial plug and connector body can be substituted.
- **Drop 1 needs one receptacle.** The UPS is the only thing plugged into it; everything else hangs off the UPS.
- **Quad box, only if the charging station is a separate table (tentative)** (4 outlets, 2ft 12/3 tail with a 5-15P that plugs into the drop's 5-15R). That's the simplest way to give a separate charging table several outlets. The BarnDoor part is assumed, not a ruling.
- **Where the charging station sits is open, and it doesn't gate the order.** Sam's current leaning (2026-09-28): drop 2 most likely feeds the second PDU in the rack (the Bay B CPPDU511), which would make the quad box unnecessary. Until Sam decides, the BOM keeps the quad box, marked tentative. If Bay B wins, drop 2's 5-15R takes the CPPDU511 cord directly and the quad box line comes off.
- Voltage drop: 12AWG over 125ft at these loads (2–3A) is about 1%. Fine.
- Plug each drop into a different wall circuit when the venue has two. If not, both on one is fine; the total is still well under 15A.

UPS side (battery-backed)

drop 1 → **CyberPower CP1500PFCRM2U** (2U, pure sine, 1000W) → **CyberPower CPS1215RM** (basic 1U strip, no surge, no meter) → Bay A gear, desk gear, lids, and the **POWER > WING** powerCON outlet → short powerCON umbilical → Wing.

- One UPS backs both cases. The BWING can't carry its own UPS, so this is the only way to back the Wing. For standalone use the Wing goes on a wall outlet. *Decided.*
- **No surge behind the UPS.** That's why the ProX X-PC10ASTRO was passed on and why the laptop strip is on the charging side. *Decided.*
- The UPS LCD meters the combined booth + Wing load.
- **Outlet count.** The CPS1215RM has 12 outlets. The UPS side has 12 loads: ATEM, HD Mini, Resi brick, LED transmitter, TL-SG1005P brick, TL-SG105 brick, AX21 brick, fan, two monitor bricks, the Hollyland RX brick and the POWER > WING feed cord. Six are wall-warts that block a neighbour. The UPS's own outlets (8 on the CP1500PFCRM2U) take the overflow; if bricks still crowd the strip, 1ft outlet-saver pigtails fix it (not in the BOM).

Charging side (not backed)

drop 2 → **Clear Power CPPDU511** (Bay B front, metered, drawer down-lights) → **Tripp Lite RS1215-RA** (Bay B rear, camera chargers), plugged into a CPPDU511 rear outlet so both read on one meter. The **laptop strip** (plain surge strip) is also on this side. *Decided.* If the charging station turns out to be a separate table, drop 2 ends in the quad box, which feeds the table's chargers and the CPPDU511 cord (tentative).

- Label both Bay B strips **"chargers + lights + laptops only"**. No high-draw loads. Keep the total under 15A.

- Laptops run on their own batteries if this side loses power.

Estimated draw, UPS side

Device	Est.
ATEM TV Studio Pro 4K	~35W
HyperDeck HD Mini	~10W
Resi	~20W
LED wall transmitter	~30W
Production switch + AX21	~20W
Dante satellite switch	~5W
UltraStudio + Hollyland RX	~15W
2 × lid monitors	~50W
Wing (over the umbilical)	~90–100W
Total	**~275–285W**

- That's 28% of the UPS's 1000W. **Spare: ~715–725W (about 6A at 120V).** Enough for Starlink (~50–75W) and other ad-hoc gear on backup.
- Estimates only. The UPS LCD shows the real number; log it at the first bench test.
- Field devices (confidence monitor, LED wall, cameras and converters) are powered where they sit.
- Utility side: chargers, drawer lights and laptops, roughly 120–180W for the laptops plus chargers. Well under 15A.

Umbilical (AVPRO ↔ Wing)

Five short cables, all color-coded and labeled. The cases always sit side by side. *Decided.*

Cable	Length	Ends	Owned?
PROD trunk	15ft	etherCON both ends	buy (BOM trunk cable + carriers)
DANTE trunk	15ft	etherCON both ends	buy (BOM trunk cable + carriers)
TALKBACK	15ft	XLR	owned (Sam)
IEM	15ft	XLR	owned (Sam)
POWER	15ft, 14/3 SJOOW	powerCON TRUE1 male (AVPRO end), female (Wing end)	buy

- The 15ft length leaves room to angle the cases.
- **Why 14/3 for the power link:** Neutrik lists the TRUE1 cable connector (NAC3FX-W) for 6–12mm cable. 12/3 SOOW (the drop cable) is 15.1mm and won't fit; 14/3 SJOOW is 9.4mm, rated 18A. The Wing draws about 1A.
- **Flexibility:** if a venue ever needs the cases apart, add longer trunks and a longer powerCON link then. Nothing on either panel changes.
- powerCON family: one matched TRUE1 family for the outlet, both cable ends, and the Wing inlet. You or the vendor confirm it before we order.

6. Fill hardware

Quantities and prices are in the attached `BOM.xlsx` (tabs: Buy, Owned, Check, Changes). The last column below is the price status (est / firm / quoted / tentative), not the price. Cases and freight are left off this list; they're question 1.

- **First order:** $4,671.84, of which $1,650.00 is the case and freight placeholder (AVPRO $500, BWING $900, freight $250). Without cases: $3,021.84.
- **Wing panel block:** $100.00, bought only if the BWING can take a panel.

First order

Group	Item	Model / SKU	Qty	Price status
AVPRO panel	1U 12-hole D-series plate	My Cable Mart D-series	3	est
AVPRO panel	BNC 75 ohm feedthrough (camera SDI)	Neutrik NBB75DFI	8	est
AVPRO panel	HDMI feedthrough	Neutrik NAHDMI-W-B	1	est
AVPRO panel	etherCON feedthrough	Neutrik NE8FDP-B	6	est
AVPRO panel	XLR female chassis	Neutrik NC3FD-L-B-1	1	est
AVPRO panel	XLR male chassis	Neutrik NC3MD-L-B-1	1	est
AVPRO panel	powerCON TRUE1 chassis outlet	Neutrik NAC3FPX (family to confirm)	1	est
AVPRO panel	Feed cord for the POWER > WING outlet	12/3 cord with 5-15P, wired to the outlet	1	est
AVPRO panel	D-hole blanking plate	Neutrik DBA-BL	18	est
Spares	BNC feedthrough, spare	Neutrik NBB75DFI	2	est
Spares	etherCON feedthrough, spare	Neutrik NE8FDP-B	2	est
Spares	XLR female chassis, spare	Neutrik NC3FD-L-B-1	1	est
Spares	XLR male chassis, spare	Neutrik NC3MD-L-B-1	1	est
Cable ends	etherCON cable carrier (standard etherCON, over an RJ45 plug)	Neutrik NE8MX-1 (to confirm)	11	est
Cable ends	RJ45 plugs, shielded (pack), for the carrier ends	Shielded Cat6A RJ45 plugs, 10-pk	1	est
Cable ends	powerCON TRUE1 cable connector, male	Neutrik NAC3MX-W (to confirm)	2	est
Cable ends	powerCON TRUE1 cable connector, female	Neutrik NAC3FX-W (to confirm)	2	est
Power	UPS, 2U pure sine (backs booth + Wing)	CyberPower CP1500PFCRM2U	1	est
Power	Rack PDU from the UPS (basic, no surge)	CyberPower CPS1215RM	1	firm

Group	Item	Model / SKU	Qty	Price status
Power	Bay B front PDU, metered, drawer lights	Clear Power CPPDU511	1	firm
Power	Bay B rear strip, camera chargers	Tripp Lite RS1215-RA	1	quoted 2026-08-20
Power	Laptop power strip, plain surge strip	Surge strip (you're sourcing)	1	est
Cooling	1U thermostatic exhaust fan	AC Infinity CLOUDPLATE T1	1	est
Cooling	1U perforated blank panel	Steel perforated (Amazon)	3	est
Mounting	1U vented cantilever shelf, 10in deep	AC Infinity 1U vented (B01C9KYUG8)	3	est
Network	Wing case Dante switch (non-PoE)	8-port metal gigabit switch (TL-SG108 or equivalent)	1	est
Storage	4U rack drawer	StarTech 4UDRAWER	2	est
Cable mgmt	1U cable lacing bar	StarTech 12S-CABLE-LACING-BAR	1	est
Cable mgmt	Hook-and-loop straps	ONE-WRAP + WrapStrap	1	est
Hardware	Rack screws + cage nuts	10-32 (assumed, verify on the rails)	1	est
Cabling	SDI BNC jumpers, panel to ATEM	Belden 1855A / premade BNC-BNC	8	est
Cabling	Cat6A patch, color-coded (lot)	Monoprice SlimRun Cat6A 10-pk	1	est
Cabling	Trunk cable, shielded Cat6A, 15ft	Shielded Cat6A patch, 15ft	2	est
Cabling	HDMI 2.0 jumper, short	HDMI 2.0, 6-10ft	3	est
Cabling	Short power cord, 5-15P to IEC C13	2-3ft	6	est
Cabling	Umbilical power cable (AVPRO > Wing), 14/3 SJOOW	14/3 SJOOW, cut to length (ft)	15	est
Venue power	Power drop cable, 12/3 SOOW (2 x 125ft)	12/3 SOOW, cut to length (ft)	250	est
Venue power	Edison plug, 5-15P (wall end)	Hubbell HBL5266C	2	est
Venue power	Edison connector body, 5-15R (rig end)	Hubbell HBL5269C	2	est
Venue power	Quad box, 4 outlets, 2ft 12/3 tail with 5-15P	BarnDoor QB-M-2FT metal quad box	1	tentative, may drop
Monitors	Lid monitor, 24in matte IPS	LG UltraGear 24GS60F-B	2	est
Monitors	Right-angle HDMI + DC adapters (lot)	90-deg HDMI + DC barrel	1	est

Wing panel block (conditional)

Group	Item	Model / SKU	Qty	Price status
Wing panel	Small D-series plate (8 holes)	D-series plate	1	est

Group	Item	Model / SKU	Qty	Price status
Wing panel	etherCON feedthrough	Neutrik NE8FDP-B	3	est
Wing panel	XLR male chassis	Neutrik NC3MD-L-B-1	1	est
Wing panel	XLR female chassis	Neutrik NC3FD-L-B-1	1	est
Wing panel	powerCON TRUE1 chassis inlet	Neutrik NAC3MPX (family to confirm)	1	est
Wing panel	Inlet-to-Wing power pigtail	IEC C13 cord, cut and wired to the inlet	1	est
Wing panel	D-hole blanking plate	Neutrik DBA-BL	2	est
Wing panel	FALLBACK ONLY: powerCON TRUE1 to IEC C13 jumper	TRUE1 male to C13, short	0	est

7. Layout

ELIM AVPRO - Rack Layout

Build set, 2026-09-28. ProX AVPRO workstation (two 12U bays, two 4U tops) + the Wing in a BWING flip case. Bay A = active system; Bay B = storage + utility power.

Camera / SDI = Blue · HDMI = White · Dante = Green · Production net = Yellow · Internet = Red · LED data = Purple · Analog XLR + AES50 = Orange · Power = Black · Blank / reserved

01 Front - operator side

Two lid monitors and the work surface above the bays. Nothing plugs into a device directly.

MONITORS - 2 lids (300 nit matte)

- **LEFT lid** — ATEM hardware multiview, day one. LG 24GS60F-B
- **RIGHT lid** — second operator view, fed in Phase 3. LG 24GS60F-B

DESK

- **ATEM TV Studio Pro 4K** — retention tray over the 3in grommet; I/O drops into Bay A
- **AVIO (Dante)** — XLR pigtails to ATEM Ch 1-2 | PoE from the satellite switch
- **UltraStudio Monitor** — behind the lids | ProPresenter USB-C > SDI > ATEM IN 8
- **Switcher laptop** — ATEM Software Control + HyperDeck + Resi
- **ProPresenter Mac (Bianca)** — HDMI > LED Tx | Anker hub > prod ETH + CONF OUT | USB-C > Dante
- **Laptop strip** — plain surge strip, fed from Bay B (utility)

BAY A - active system (front)

4U top	Hollyland RX only (SDI direct to ATEM IN 2)
U12	**Fan - 1U thermostatic** — AC Infinity CLOUDPLATE T1; quiet until warm
U11	**Resi** — 1U vented shelf
U10	**HD Mini + Dante satellite (TL-SG1005P)** — 1U vented shelf (+ switch power brick)
U9	**LED wall transmitter** — rack-mounted
U8	**Production switch (TL-SG105) + AX21** — 1U vented shelf (antennas up/out)
U7	**Cable management** — lacing bar
U6-4	**Vented blanks (reserve)** — airflow + growth
U3	**CPS1215RM strip (from UPS)** — basic 1U, no surge, no meter
U2-1	**UPS - CP1500PFCRM2U (2U)** — pure sine; backs booth + Wing

BAY B - storage + utility power (front)

4U top	reserved
U12	**CPPDU511 front / RS1215-RA rear** — CPPDU511: metered, drawer lights; fed from drop 2
U11-9	**Reserved** — bigger switch or Mac Minis
U8-5	**4U drawer (upper)** — storage
U4-1	**4U drawer (lower)** — heavier items here to balance the case

02 Rear - cable side

Field cables land on the rear I/O panel. Volunteers work from this face. Rear notes show what connects at each device.

BAY A - rear

4U top	Hollyland RX only (SDI direct to ATEM IN 2)
U12	**Fan - 1U thermostatic** — exhausts toward the open back
U11	**Resi** — SDI in < ATEM OUT 8 | ETH > AX21
U10	**HD Mini + Dante satellite (TL-SG1005P)** — HD Mini SDI < ATEM AUX | switch: DANTE TRUNK, AVIO (PoE), Bianca Dante

BAY B - rear

4U top	reserved
U12	**CPPDU511 front / RS1215-RA rear** — RS1215-RA: camera chargers, plugs into CPPDU511
U11-9	**Reserved**

U9	**LED wall transmitter** HDMI < ProPresenter | 2x ETH > LED 1, LED 2
U8	**Production switch (TL-SG105) + AX21** PROD TRUNK, INTERNET IN, SPARE land here
U7	**Cable management** REAR I/O PANEL (3U) behind the upper block
U6-4	**Vented blanks (reserve)**
U3	**CPS1215RM strip (from UPS)** UPS-backed gear + POWER > WING feed
U2-1	**UPS - CP1500PFCRM2U (2U)** line cord to power drop 1; LCD meters ~275-285W

U8-5	**4U drawer (upper)**
U4-1	**4U drawer (lower)**

03 Rear I/O panel (3U, three plates)

Neutrik D-series. Panel 1 video, panel 2 data / audio / power, panel 3 spare. * = shared jack (see notes).

PANEL 1 - VIDEO

CAM 1	CAM 2 *	CAM 3	CAM 4	CAM 5	CAM 6	CAM 7	CAM 8 *	CONF OUT	blank	blank	blank
BNC	BNC	BNC	BNC	BNC	BNC	BNC	BNC	HDMI	-	-	-

PANEL 2 - DATA / AUDIO / POWER

DANTE TRUNK	LED 1	LED 2	INTERNET IN	PROD TRUNK	TALKBACK	IEM	SPARE	blank	blank	blank	POWER > WING
etherCON	etherCON	etherCON	etherCON	etherCON	XLR-F	XLR-M	etherCON	-	-	-	powerCON

PANEL 3 - SPARE (expansion)

spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row	spare row
-	-	-	-	-	-	-	-	-	-	-	-

04 Wing panel (planned, pending BWING inspection)

Five umbilical jacks to the AVPRO + one AES50 to the stage box. Genders mirror the AVPRO panel. If the BWING can't take a panel, the umbilical plugs straight into the Wing.

WING PANEL

CONTROL	DANTE TRUNK	AES50	TALKBACK	IEM	POWER IN	blank	blank
etherCON	etherCON	etherCON	XLR-M	XLR-F	powerCON	-	-

Inside the case: 8-port Dante switch (TL-SG108 or equivalent) + Tyler's Mac (USB-C to the Wing, plus Dante).

05 Notes

1. Colors = the cable color code. Every cable also carries a number and a source > destination label: match the color, match the number.
2. Bay A is the whole active system. Every network jumper stays in Bay A; nothing crosses bays. Bay B is storage and utility power; no signal.
3. * Shared jacks: the Hollyland RX and the UltraStudio plug straight into ATEM IN 2 and IN 8. The CAM 2 and CAM 8 jumpers are parked beside them; swap one in to use that jack for a field camera.
4. Power: two 125ft 12/3 SOOW drops from wall outlets (5-15P > 5-15R). Drop 1 feeds the UPS, which backs the booth and the Wing (~275-285W of 1000W, ~715-725W spare). Drop 2 feeds the charging station (most likely the Bay B PDU; a quad box only if it's a separate table); the Bay B strips (CPPDU511 + RS1215-RA) and the laptop strip are on that side. No surge behind the UPS.
5. Cooling: the open back is primary. The fan exhausts toward the back. Reserve U's are vented blanks. Keep the back clear at venues.
6. Umbilical = 5 short cables (15ft; the cases sit side by side): PROD, DANTE, TALKBACK, IEM and POWER (powerCON TRUE1 on 14/3 SJOOW). AES50 is stage-side.
7. Wing panel is planned, pending inspection of the BWING at delivery. Without it, the umbilical plugs straight into the Wing's own jacks.
8. Dimensions: 12U bays 21.65in deep; 4U tops 4.33in; work surface 49.6 x 16.9in. Side compartments are not used in the design (confirm on delivery).
9. Monitors: 2x LG 24GS60F-B, 300 nit matte, fine under a tent or canopy; direct sun needs angling. Right-angle HDMI/DC adapters for lid clearance.